Evaluating 30-Day Hospital Readmission Prediction in Patients with Diabetes: Patient-Level Versus Encounter-Level Validation

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***Abstract*—Performance estimates for readmission prediction depend on whether repeated encounters from the same patient can cross evaluation partitions. We investigate this dependence using the Diabetes 130-US Hospitals dataset. A reproducible audit identifies 99,343 eligible encounters from 69,990 patients after excluding death and hospice discharges; 11.39% have the positive readmission label. We compare L2 and L1 logistic regression, random forest, and LightGBM under patient-disjoint and stratified encounter-level evaluation. Each protocol is repeated across five predefined seeds, with five-fold development-set validation and training-only preprocessing. Average precision is the primary outcome. The four weighted models achieve mean patient-level AP between 0.208 and 0.228; LightGBM has the highest mean. Encounter-level test sets contain 41.7% encounters from patients represented in development data on average. Encounter-level mean AP is modestly higher (+0.005 to +0.013), with a change in model ranking. These initial experiments quantify evaluation sensitivity without treating a difference between distinct test populations as causal evidence of leakage. Calibration correction, resampling, and detailed subgroup analyses remain planned.**

***Index Terms*—readmission, diabetes, grouped validation, repeated encounters, average precision**

I. Introduction

Readmission within 30 days can increase healthcare costs and indicate difficulties with recovery, discharge planning, or follow-up care [1]. Patients with diabetes often have multiple diagnoses and repeated admissions, making risk assessment challenging. At discharge, identifying higher-risk encounters could help prioritize follow-up resources. Credible evaluation is therefore necessary before apparent predictive gains can inform care. We use historical clinical records to assess both prediction and the reliability of its evaluation.

The Diabetes 130-US Hospitals dataset contains inpatient encounters collected during 1999–2008 [2], [3]. An encounter is not an independent patient: individuals may contribute several records. An encounter-level random split can therefore evaluate a mixture of previously represented and unseen patients, whereas a patient-level split evaluates individuals absent from development data. Removing patient identifiers from predictors does not remove this dependence, because clinical and utilization characteristics may recur across visits. Conversely, the presence of overlap alone does not quantify its effect on performance.

Prior studies examine clinical associations [3], predictive models [1], and patient-level safeguards [4]. We extend this work through repeated protocol comparisons on a fixed cohort.

Our questions are: **RQ1**, how well do four established models rank positive encounters under patient-disjoint evaluation? **RQ2**, how do average precision and model rankings change under encounter-level evaluation across seeds? **RQ3**, how much patient overlap occurs, and how do probability quality and performance within seen/unseen patient subsets qualify the comparison? Prediction is defined at discharge, not at admission; the main target is generalization to unseen patients within this historical dataset.

The specific contributions of this report are (1) a verified cohort and repeated-patient audit, including separate treatment of unperformed laboratory tests; (2) a reproducible comparison of four model families under two validation protocols and five seeds; and (3) an uncertainty-aware interpretation of overlap, ranking, and calibration. The target audience remains the IEEE Intermountain Engineering, Technology, and Computing Conference.

II. Related Work

A. Clinical analysis and predictive benchmarking

Strack et al. [3] restricted their analysis to the first encounter per patient and excluded death/hospice outcomes. This avoids repeated observations by changing the cohort and supports their analysis of HbA1c testing and readmission. Our study retains repeated encounters to investigate evaluation dependence explicitly. The two designs address different questions, so their sample size and reported associations are not interchangeable with our predictive results.

Emi-Johnson and Nkrumah [1] compare logistic regression, random forest, XGBoost, and a neural network using a stratified 80/20 split, class weighting, cross-validation, and SHAP. Patient-disjoint partitioning is not explicitly established in the reported splitting procedure. That is a reporting distinction, not proof that their implementation leaked information. Their work

motivates model comparison, while our emphasis is partition sensitivity rather than cross-paper score superiority.

B. Leakage safeguards and the closest comparison

Afridi et al. [4] explicitly describe patient-level partitioning, fold-specific preprocessing and feature selection, SMOTENC/class weighting, and SHAP/LIME explanations. Their analytical cohort contains 75,081 encounters, and their partition/training description specifies seed 42. Our comparison adds repeated protocol evaluation, measured overlap and ranking stability on a fixed cohort (Table I).

C. Validation matched to the intended use

Saeb et al. [5] show why validation must reflect the clinical use case: records from the same subject can produce overly optimistic estimates for new-subject prediction. Kapoor and Narayanan [6] place such problems within a broader taxonomy of leakage and emphasize transparent computational evidence. Together, these studies motivate our grouped primary evaluation and explicit overlap diagnostic. Our contribution applies these principles to repeated diabetic encounters while retaining a record-level comparison to measure sensitivity. Random record partitioning also lacks the chronology needed to evaluate future visits from known patients.

D. Imbalance and evaluation

Precision–recall analysis is useful when positives are uncommon [7]. We operationalize it as non-interpolated average precision (AP), rather than trapezoidal PR area [8]. A constant-score reference makes test prevalence visible. Group-aware validation addresses dependence between observations from the same person [9]. For imbalance, weighting and resampling are distinct choices: both can change the meaning of fitted probabilities, so ranking performance alone is insufficient. We include an unweighted logistic reference and Brier scores; SMOTENC through imbalanced-learn [10] remains a planned comparison. LightGBM [11] supplies a computationally feasible nonlinear model alongside random forest and regularized logistic regression.

III. Data and Preprocessing

A. Source, cohort, and outcome

We downloaded the public UCI files [2], including their categorical-ID mapping, and recorded a SHA-256 checksum. The CSV has 101,766 rows and 50 columns: two identifiers, the outcome, and 47 candidate predictors. The unit of analysis is one inpatient encounter. The outcome is $y = 1$ for `<30`, and $y = 0$ for `>30` or `NO`; no additional readmission labels were inferred.

Fixed eligibility rules exclude discharge codes 11, 13, 14, 19, 20, and 21 (death or hospice categories in the supplied mapping). They remove 2,423 rows, leaving 99,343 encounters, 69,990 patients, and 11,314 positives (11.39%). All age and gender categories are retained. Transfers remain eligible in this version, with their discharge category represented as a predictor. This decision requires a later sensitivity analysis because transfers need not represent an ordinary community discharge.

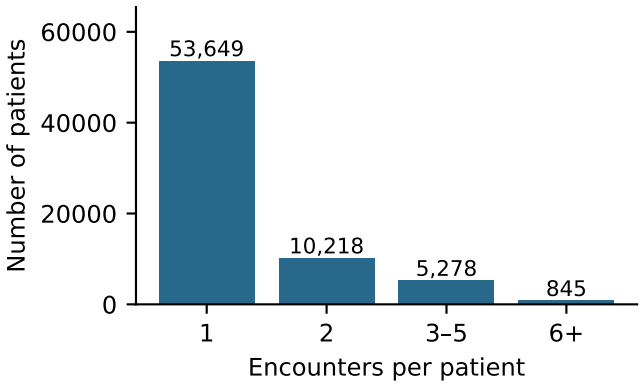


Fig. 1. Eligible patients grouped by encounter count. Repeat encounters are retained and grouped during primary evaluation.

B. Completed quality audit

The original data contain zero exact duplicate rows, zero repeated encounter IDs, and zero duplicate rows after removing the two identifiers. Repeated patients are a separate phenomenon: 16,341 patients (23.35% of eligible patients) contribute 45,694 encounters (46.00% of eligible encounters). The median is one encounter, the 90th percentile two, the 95th three, the 99th six, and the maximum 40. Figure 1 shows the distribution. These observations support preserving patient groups rather than deleting repeat visits.

Table II separates structural test absence from unknown values. In particular, `None` in HbA1c/glucose fields is read literally and recoded as "Not performed," consistent with the dataset definitions [3]; it is not numerically imputed. The `?` marker becomes an explicit "Unknown" category. Weight is excluded according to the pre-existing plan. Unknown diagnosis counts are 20, 356, and 1,419 for the three diagnosis fields; they remain explicit unknown categories. Numerical ranges and category counts are exported for review. No outcome-dependent outlier deletion or winsorization is performed.

C. Representation and safeguards

The model input has 46 variables after removing weight and both IDs. Eight count/duration variables are standardized using training-fold means and standard deviations. The remaining variables, including age bands and admission/discharge IDs, are categorical. ICD-9 diagnoses are mapped by fixed rules into nine clinical groups plus unknown, adapting the grouping in [3] to all three diagnosis fields. The mapping treats supplemental E/V codes as other and preserves unknown codes separately. Categorical vocabulary is learned through training-only one-hot encoding; unseen categories are ignored at transformation time. No numerical missing values require imputation in this cohort.

Patient and encounter IDs are used for grouping, assertions, and saved split manifests only. Fixed eligibility rules and

TABLE I
COMPARISON OF RELEVANT STUDIES AND THIS REPORT. MISSING REPORTING IS NOT EVIDENCE OF AN INCORRECT IMPLEMENTATION.

Study	Repeated patients	Validation	Leakage/imbalance handling	Relation to our study
Strack et al. [3]	First encounter per patient retained	Association modeling; not the same predictive holdout comparison	Death/hospice exclusions; clinical covariate analysis	Dataset semantics and cohort rationale
Emi-Johnson and Nkrumah [1]	Patient grouping not explicit in split description	Stratified 80/20; five-fold model tuning	Preprocessing pipeline and class weighting reported	Conventional predictive benchmark
Afridi et al. [4]	Patient groups kept separate	Patient holdout; five-fold grouped validation; seed 42 specified	Training-fold preprocessing, selection and SMOTENC/weights	Closest framework; safeguards already established
This report	All eligible encounters retained	Two protocols, five seeds; matched search budgets	Training-only fitting; overlap assertions; no test resampling	Quantifies protocol sensitivity and ranking variability

TABLE II
COMPLETED AUDIT OF THE ELIGIBLE COHORT. UNKNOWN VALUES ARE DISTINCT FROM TESTS NOT PERFORMED.

Item	Count (percentage)
Eligible encounters / patients	99,343 / 69,990
Positive / negative outcomes	11,314 / 88,029
Patients with repeat encounters	16,341 (23.35%)
Encounters from repeat patients	45,694 (46.00%)
Unknown weight	96,218 (96.85%)
Unknown medical specialty	48,616 (48.94%)
Unknown payer code	39,398 (39.66%)
Unknown race	2,234 (2.25%)
HbA1c not performed	82,509 (83.05%)
Glucose serum test not performed	94,191 (94.81%)

semantic mappings precede splitting; all learned operations follow it (Fig. 2). No longitudinal features are reconstructed from future encounters. The supplied prior-utilization variables are used as documented. Because usable encounter dates and hospital group identifiers are unavailable, patient-disjoint validation does not establish temporal or new-hospital generalization.

IV. METHODOLOGY AND IMPLEMENTATION

A. Models and objectives

A constant predictor outputs the development-set prevalence. L2 logistic regression is the main model baseline; L1 logistic regression evaluates sparse coefficient selection through a LASSO-type penalty [12]. For both, class weights are $w_k = n/(2n_k)$ using only the current training labels. Logistic models minimize a regularized weighted binary log loss, with inverse penalty strength C selected from $\{0.1, 1\}$ for L2 and $\{0.01, 0.1\}$ for L1. L1 and L2 penalties differ in whether coefficients are encouraged toward exact zero or smoothly shrunk. We use the liblinear solver and a 1,000-iteration limit. An otherwise equivalent unweighted L2 model provides a probability-quality reference.

Random forest [13] aggregates 100 trees with maximum depth 12 and square-root feature subsampling; minimum leaf size is selected from $\{5, 20\}$. LightGBM uses 150 boosting rounds, learning rate 0.05, minimum leaf count 50, and leaves selected from $\{15, 31\}$. Both use training-derived balanced class weights. These deliberately small, prespecified searches control runtime and provide equal two-candidate budgets rather than an exhaustive benchmark. No early stopping uses test data. Model configurations and selected values are exported per run.

B. Completed versus planned components

The data audit, semantic preprocessing, sparse encoding/scaling, four weighted models, unweighted logistic reference, two splitting protocols, repeated tuning/evaluation, and output generation are implemented. L1 performs embedded feature selection; no separate RFE step has been run. SMOTENC, post-hoc probability recalibration, SHAP/permutation interpretation, and discharge/subgroup ablations remain planned. Thus their benefits are not asserted in this report.

The completed run used Python 3.12.14, scikit-learn 1.8.0 and LightGBM 4.7.0. The environment uses Python, NumPy, pandas, SciPy, scikit-learn, LightGBM, and Matplotlib [14], [15], [16], [17], [11], [18]. Versions and the checksum are recorded. Sparse encoding reduces memory use; random forest receives a dense float32 matrix for faster fitting. Model libraries are limited to two threads. A timing pilot narrowed the L1 search before test scoring. The five seed jobs ran concurrently with two threads per model and an eight-core CPU quota. Logged outer-partition runtimes were 266–367 seconds (median 314 seconds); the longest seed job took 712 seconds for both protocols, approximately 12 minutes excluding setup and analysis. These are shared-resource wall times, not isolated model benchmarks. The repository provides code, aggregate results and manuscript source; encounter-level predictions and split manifests are generated locally:
https://github.com/Ahmadjajja/Hospital_Readmission_Prediction.

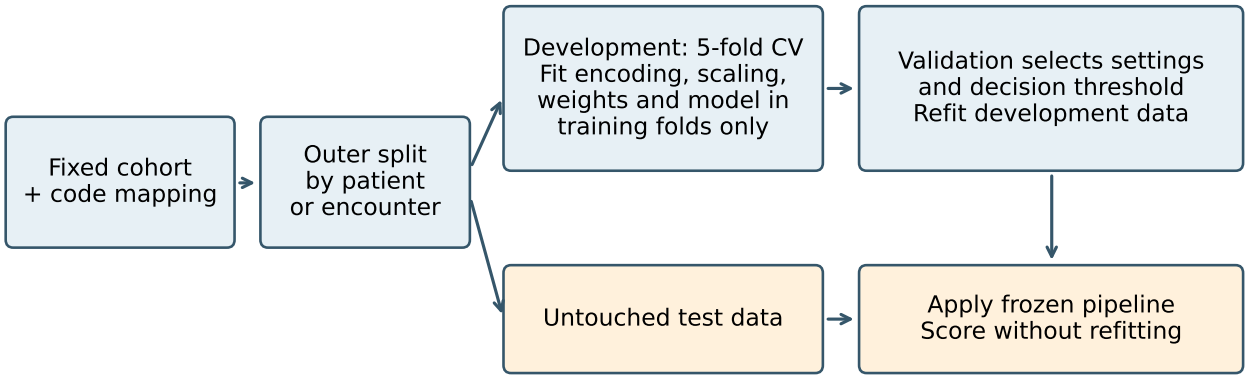


Fig. 2. Implemented workflow. The test branch bypasses all fitting and selection. L1 selection occurs inside training folds. SMOTENC is planned, not used in the reported runs; when added it will be confined to training folds before final encoding.

V. Experimental Design

A. Repeated outer splits and internal selection

For each seed in $\{11, 23, 42, 67, 101\}$, GroupShuffleSplit holds out 20% of unique patients. All encounters of those patients enter the test set, so the fraction of encounters is only approximately 20%. The comparison uses stratified random splitting of 20% of encounters. Each arm independently creates five development folds: StratifiedGroupKFold for the patient arm and StratifiedKFold for the encounter arm. Assertions verify zero patient intersection in every primary outer and inner split.

Each candidate is evaluated by mean fold AP. The selected candidate is refitted on the full development set. The F1-maximizing threshold is chosen from its pooled out-of-fold development predictions, then frozen for the test evaluation. The same candidate sets and preprocessing rules are used across arms, but independently selected parameters may differ. This estimates the effect of the complete validation protocol rather than a pure causal effect of outer-split overlap. Test results do not feed back into tuning.

B. Metrics and uncertainty

Average precision is computed as

$$\mathrm{AP} = \sum_j (R_j - R_{j-1}) P_j, \tag{1}$$

where P_j and R_j are precision and recall along the score-ranked curve [8]. We also report ROC-AUC, precision, recall, F1, and Brier score $n^{-1}\sum_i (p_i - y_i)^2$. Threshold tuning affects the threshold-dependent measures, not AP or ROC-AUC. Accuracy is not a success criterion because an always-negative classifier exploits the imbalance.

Across seeds, we report means and sample standard deviations. The repetitions reuse patients and are not independent population samples; their standard deviation describes partition/model variability, not a confidence interval for clinical deployment. For a prespecified illustrative patient holdout (seed 42), 1,000 patient-cluster bootstrap samples (bootstrap RNG seed 2026) estimate percentile 95% intervals for AP differences against L2. All encounters of a sampled patient receive the same multiplicity, and all models use identical sampled clusters. These intervals are conditional on fitted models and that holdout; they do not incorporate all training uncertainty or adjust for multiple comparisons. We do not apply paired intervals across distinct protocol test sets.

C. Overlap and success criteria

For each encounter-split test set, we measure the fraction of test encounters whose patient appears in development data and count overlapping patients. We separately summarize seen/unseen subsets, including prevalence and sample size. These subsets differ in case mix and repeat frequency, so their differences are descriptive. A credible outcome is a complete, reproducible comparison with uncertainty, even when a complex model does not improve on the baseline or the split gap is small. Evidence of predictive improvement requires both effect size and uncertainty, not a predetermined score target.

VI. Implementation Progress and Initial Findings

A. Completed experiments and performance

All ten outer partitions completed: 500 candidate fits, 50 selected model fits, and ten constant-reference evaluations. The exported metrics contain 60 evaluated model/protocol/seed combinations. Automated checks recompute metrics from predictions, verify labels against source records, and verify partition integrity. No convergence warnings were observed in the completed-run logs. Table III and Fig. 3 summarize the main results. Values are preliminary benchmark findings, not clinical validation.

LightGBM has the highest mean patient-level AP (0.228), compared with 0.216 for weighted L2 and 0.112 for the constant reference. The encounter-minus-patient differences in mean AP are +0.0055 (L2 logistic), +0.0062 (L1 logistic), +0.0132 (Random forest), +0.0102 (LightGBM). These are

TABLE III
TEST PERFORMANCE, MEAN ± SAMPLE SD ACROSS FIVE SEEDS. CONSTANT AP EQUALS EACH TEST SET'S POSITIVE FRACTION.

Model	Split	AP	ROC-AUC
L2 logistic	Patient	0.216 ± 0.006	0.663 ± 0.003
L2 logistic	Encounter	0.221 ± 0.002	0.669 ± 0.005
L1 logistic	Patient	0.215 ± 0.006	0.665 ± 0.003
L1 logistic	Encounter	0.222 ± 0.002	0.671 ± 0.004
Random forest	Patient	0.208 ± 0.002	0.665 ± 0.005
Random forest	Encounter	0.221 ± 0.002	0.673 ± 0.004
LightGBM	Patient	0.228 ± 0.006	0.673 ± 0.002
LightGBM	Encounter	0.238 ± 0.004	0.679 ± 0.003
L2 unweighted	Patient	0.216 ± 0.006	0.662 ± 0.003
L2 unweighted	Encounter	0.221 ± 0.002	0.668 ± 0.005
Constant	Patient	0.112 ± 0.002	0.500 ± 0.000
Constant	Encounter	0.114 ± 0.000	0.500 ± 0.000

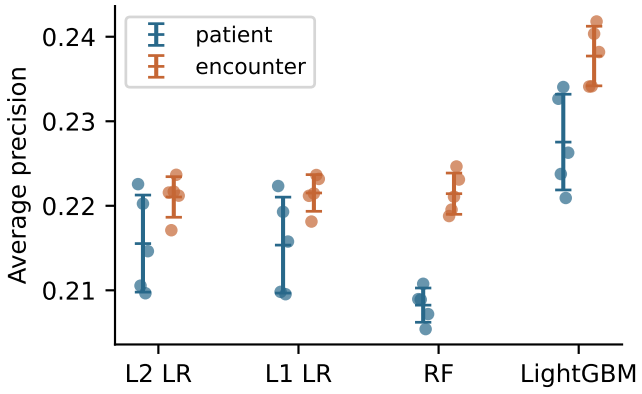


Fig. 3. AP across five seeds. Points are individual runs; marks and error bars show means and sample standard deviations.

descriptive differences across separately constructed test sets. They are not paired causal effects of overlap.

B. Overlap and conditional uncertainty

Patient-level holdouts have zero patient overlap by assertion. Encounter-level overlap ranges from 41.34% to 42.10% of test encounters (mean 41.70%). Patient-arm test prevalence ranges from 10.99% to 11.41%; the corresponding test sizes range from 19,717 to 19,991. The constant-reference results make this variation explicit.

For LightGBM within encounter-level holdouts, mean AP is 0.279 for seen patients and 0.157 for unseen patients. Their mean positive fractions are 20.10% and 5.16%, respectively. The case-mix and prevalence difference prevents interpreting this subset contrast as an isolated overlap effect.

For the prespecified patient holdout at seed 42, paired AP differences relative to weighted L2 and their 95% patient-bootstrap intervals are L1 logistic: -0.0010 [-0.0026, +0.0006]; Random forest: -0.0095 [-0.0169, -0.0021]; LightGBM: +0.0138 [+0.0073, +0.0214]. These conditional, unadjusted intervals complement rather than replace the repeated-seed summaries.

TABLE IV
PATIENT-LEVEL THRESHOLD AND PROBABILITY DIAGNOSTICS: MEANS ACROSS FIVE RUNS. THRESHOLDS MAXIMIZE DEVELOPMENT OUT-OF-FOLD F1.

Model	Precision	Recall	F1	Brier
L2 logistic	0.193	0.475	0.274	0.225
L1 logistic	0.193	0.476	0.275	0.225
Random forest	0.189	0.500	0.274	0.219
LightGBM	0.211	0.435	0.284	0.215
L2 unweighted	0.197	0.447	0.272	0.095
Constant	0.000	0.000	0.000	0.099

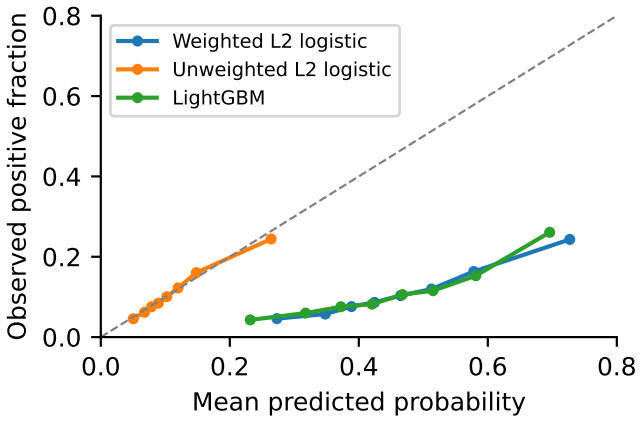


Fig. 4. Calibration diagnostic for the prespecified patient holdout, seed 42. Eight quantile bins; dashed line indicates ideal calibration. This is a diagnostic, not a fitted correction.

C. Thresholds and probability quality

Table IV reports the precision–recall trade-off after development-only threshold selection. Unweighted L2 has mean Brier score 0.095, compared with 0.225 for weighted L2. The calibration diagnostic (Fig. 4) illustrates why class-weighted scores should not automatically be interpreted as calibrated risks. Brier score combines discrimination and calibration; a lower value alone does not establish perfect calibration. No probability correction has yet been fitted.

Under the Report 1 decision rule, LightGBM improves on weighted L2 in the seed-42 patient holdout because its AP-difference interval excludes zero; random forest is worse and L1 is inconclusive. This inference is conditional on that holdout. Across seeds, the encounter-minus-patient gap is modest (+0.005 to +0.013 AP). Random forest ranks last under patient evaluation (0.208 AP), but approximately ties L2 under encounter evaluation (both round to 0.221). Weighted models have Brier scores near 0.22, worse than the constant reference (about 0.099), reinforcing the need for probability calibration before interpreting their outputs as risk.

VII. DISCUSSION, LIMITATIONS, AND RESPONSIBLE USE

Patient-level validation addresses unseen patients; encounter-level validation mixes seen and unseen patients. Without chronology, neither simulates prospective

deployment, and their difference does not isolate a causal overlap effect.

Cohort definitions differ from prior studies, including removal of first-encounter restrictions and retention of transfer discharges. Cross-paper performance rankings would therefore be misleading. The limited search spaces may favor some models, and repeated encounters give frequent users greater weight in encounter-level metrics. A patient-weighted sensitivity analysis is planned. Out-of-system readmissions may be unobserved, and historical practice, coding, and treatment patterns may not represent current care.

UCI distributes the dataset under CC BY 4.0 [2]. The data are deidentified, but demographic fields remain sensitive; we make no reidentification attempt. The repository documents the source and license. Race, gender, and age categories may reflect uneven representation and measurement; subgroup calibration and error analysis remain unfinished. No fairness, causal, intervention-benefit, or clinical-readiness claim is made. Feature importance, when added, will describe model associations rather than treatment effects.

VIII. Response to Feedback and Updated Outline

The workflow in Report 1, Fig. 1, is now Fig. 2; it is enlarged to full text width with larger readable labels and a separate untouched-test branch. Learned preprocessing and L1 selection appear after splitting; planned resampling is explicitly training-only. Five predefined seeds and variability summaries address partition stability. Table I supplies the requested literature comparison and incorporates Afridi et al., distinguishing our protocol-comparison objective from their framework. The duplicate and encounter-frequency audit is now completed, and Table II distinguishes unperformed tests from unknown values. Completed and planned components are labeled, and references use a consistent numbered IEEE style. The research question and dataset remain unchanged; an unweighted logistic reference was added to clarify calibration effects.

Table V updates the complete manuscript outline and identifies evidence still needed.

TABLE V
Updated final-paper outline and evidence requirements.

Section	Content, evidence, and planned outputs
Abstract	Update preliminary findings after remaining analyses.
Introduction	Stable RQs and contributions; qualify scope with final evidence.
Related work	Compare patient/record validation, leakage and calibration; extend Table I.
Data	Verified cohort and missingness; retain audit table and encounter-frequency figure.
Methods	Add training-only SMOTENC and calibration configurations.
Experiments	Preserve seeds and grouping; specify sensitivity and subgroup protocols.
Results	Existing AP/ROC tables; add imbalance/ablation table, subgroup table and feature-importance figure.
Discussion	Interpret ranking changes, uncertainty and probability quality across protocols.
Limitations	Historical cohort, dependence, missing chronology, representation and external validity.
Conclusion	Answer all RQs using completed evidence.
References	Verify citations and retain consistent IEEE entries.

IX. Remaining Work and Responsibilities

Table VI prioritizes the remaining work toward Progress Report 3. Ahmad coordinates implementation and reproducibility; Muhammad Shehzore Hashmi coordinates validation and statistical checks; Akshay Kumar Dosapati coordinates literature synthesis and figures. All members will review the code, source papers, model outputs, and manuscript.

TABLE VI
Remaining work toward Report 3; prioritized internal work windows after Report 2.

Target	Task and dependency	Coordinator
Week 1	Verify runs; review person-versus-record and calibration literature; extend comparison table	Shehzore, all
Week 1–2	Training-only SMOTENC comparison on a limited model set; requires mixed-type pipeline	Ahmad
Week 2	Calibration correction from development data; discharge and patient-weighted sensitivity	Ahmad, Shehzore
Week 2–3	Subgroup counts/errors and feature-importance figure; sufficient positives required	Akshay, all
Before R3	Integrate ablation/subgroup tables, calibration and importance figures; final review	All members

If resources are limited, prioritize one-model resampling and calibration, then subgroup/interpretation analyses. Report small subgroup counts without unstable comparisons. Document protocol changes and label later reuse of current test sets as exploratory.

X. Preliminary Conclusion

Repeated-patient dependence is substantial, and encounter-level validation yields modestly higher AP and a different model ranking. LightGBM leads patient-level evaluation, while weighted probability estimates require further calibration analysis. These historical benchmark findings motivate the remaining sensitivity and interpretation work.

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